

Dr. Allen Patnaik

Deep Learning | Computer Vision | Transformers | Generative AI

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PROFESSIONAL SUMMARY

Researcher with **5+ years of hands-on experience** developing deep-learning solutions for computer vision, medical imaging, remote sensing, and generative AI applications. Ph.D. in Signal Processing and Machine Learning with expertise in **CNNs, GANs, Transformers, attention mechanisms, knowledge distillation, active learning, and image processing**. Strong programming experience in **Python, C/C++, and MATLAB**, with hands-on experience in **PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, and Git**. Experienced in end-to-end ML pipeline development, rapid prototyping, model experimentation, data preprocessing, model evaluation, and human-in-the-loop AI systems. Specialized training in Generative AI and Large Language Models.

TECHNICAL SKILLS

AI/ML	Deep Learning, Machine Learning, Computer Vision, Generative AI, CNNs, GANs, Transformers, Attention Mechanisms, Knowledge Distillation, Active Learning, Human-in-the-Loop Learning
Programming	Python, C, C++, MATLAB
Frameworks	PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn
AI Architectures	CNN, GAN, Transformer, Attention-based Networks, Residual Networks, Knowledge-Distillation Architectures
Data	Large-scale Image Data, Remote Sensing, Multispectral/Hyperspectral Data, Medical CT Imaging, Image Preprocessing, Segmentation, Feature Extraction
Engineering	Git, Linux/Unix Environments, LaTeX, ML Experimentation, Model Evaluation

PROFESSIONAL EXPERIENCE

TANUH, AI CoE – Indian Institute of Science (IISc)

Research Staff Member

Bangalore, India

Jan 2026 – Jun 2026

- Designed and developed an **AI-assisted retinal disease diagnosis system** to support clinical review and AI-assisted decision making.

Paraxial Technologies

Data Scientist

Remote

Sep 2025 – Nov 2025

- Developed an **AI-based lung cancer screening pipeline** using CT scans for automated detection and analysis of pulmonary abnormalities.
- Implemented deep-learning workflows for **CT image preprocessing, segmentation, feature extraction, and model development** using Python and PyTorch.

Indian Institute of Technology Guwahati

Teaching Assistant

Assam, India

Jan 2019 – May 2024

- Supported undergraduate and graduate courses in **machine learning, deep learning, computer vision, and image processing**.
- Conducted tutorials, technical discussions, and hands-on sessions covering practical ML/DL concepts and implementation.
- Guided students in understanding machine-learning workflows, computer-vision methods, and deep-learning architectures.

Silicon Institute of Technology

Assistant Professor

Odisha, India

Aug 2018 – Dec 2018

- Taught Digital Signal Processing, Signal Systems, and Analog Electronic Circuits at undergraduate and graduate levels.
- Conducted laboratory sessions focused on practical implementation and analysis of signal-processing concepts.

SELECTED AI/ML PROJECTS

Single Image Super-Resolution for Remote Sensing Images

2022

- Developed deep-learning models for enhancing the spatial resolution of satellite imagery.
- Designed and evaluated advanced **CNN-, attention-, adversarial-, and transformer-based architectures**.
- Investigated model architectures, preprocessing strategies, and data-augmentation techniques to improve image reconstruction quality and computational efficiency.

- Developed a **knowledge-distillation-based deep-learning framework** for long-term NDVI reconstruction using Landsat and Sentinel imagery.
- Investigated teacher-student architectures for transferring model knowledge while reducing model complexity.

Land Surface Temperature Enhancement

2023-2024

- Developed machine-learning and deep-learning models for downscaling Landsat imagery in riparian regions of Assam, India.
- Worked on geospatial data preprocessing, model development, evaluation, and spatio-temporal analysis in collaboration with Civil Engineering researchers.

AI/ML RESEARCH & ENGINEERING EXPERTISE

- Developed deep-learning architectures using **CNNs, GANs, Transformers, attention mechanisms, adversarial learning, and knowledge distillation**.
- Worked with **multispectral and hyperspectral imagery**, addressing spatial and spectral resolution enhancement.
- Investigated model architecture, preprocessing, augmentation, and evaluation strategies for improving AI model performance and efficiency.
- Developed AI solutions across **remote sensing and medical imaging**, demonstrating transfer of deep-learning methodologies across application domains.
- Research experience in **efficient model design and knowledge distillation**, relevant to resource-constrained AI deployment.

EDUCATION

Indian Institute of Technology Guwahati <i>Ph.D. – Signal Processing and Machine Learning</i>	Assam, India <i>2019 – 2025</i>
Veer Surendra Sai University of Technology <i>M.Tech.</i>	Odisha, India <i>2018</i>
Gandhi Institute for Technology <i>B.Tech.</i>	Odisha, India <i>2014</i>

SELECTED PUBLICATIONS

- **A. Patnaik**, M.K. Bhuyan and K.F. MacDorman, “A Two-Branch Multiscale Residual Attention Network for Single Image Super-Resolution in Remote Sensing Imagery,” *IEEE JSTARS*, 2024.
- **A. Patnaik**, N. Chaudhary, M.K. Bhuyan, S. Alfarhood and M. Safran, “Remote Sensing Single-Image Super-Resolution Using Convolutional Block Attention Residual Network With Joint Adversarial Mechanisms,” *IEEE Access*, 2024.
- **A. Patnaik**, A. Gour, M.K. Bhuyan and K.F. MacDorman, “Super-Resolution of Remote Sensing Image via Parallel Lattice Attention Network,” *IEEE JSTARS*, 2026.
- **A. Patnaik**, M.K. Bhuyan, S. Alfarhood and M. Safran, “Hyperspectral Image Super-Resolution via Grouped Second-Order Spatial-Spectral Attention Network,” *IEEE JSTARS*, 2025.
- **A. Patnaik**, B. Joshi, D. Sarma, M.K. Bhuyan, S. Alfarhood and M. Safran, “A knowledge distillation deep channel attention super-resolution network for long-term NDVI reconstruction using Landsat 8 and Sentinel-2 imagery,” *RSASE*, 2026.

CERTIFICATIONS & TRAINING

Generative AI with Large Language Models – Coursera	2025
Crash Course in Deep Learning with Google TensorFlow – Udemy	2024
Fundamentals of Deep Learning – NVIDIA	2021
Machine Learning for Signal and Image Processing Applications – NIT Rourkela	2018

TECHNICAL ENGAGEMENT

- Session Speaker – Convolutional Neural Networks, CDAC Program, 2025.
- Reviewer – IEEE Geoscience and Remote Sensing Letters and IEEE-GCON Conference.
- Teaching Assistant – National Programme on Technology Enhanced Learning (NPTEL).